

N'DJAMENA INTL, Chad

WMO: 647000

Lat: 12.134N

Lon: 15.034E

Elev: 968

StdP: 14.19

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	55.9	58.8	12.3	10.6	84.3	16.0	12.8	83.6	22.0	75.3	19.7	75.0	5.7	40	0.445

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	27.0	109.6	70.9	107.9	71.0	106.1	70.7	82.9	94.0	81.7	92.7	80.6	91.7	8.6	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.5	164.4	87.9	79.0	156.1	86.7	78.3	152.6	86.0	47.7	93.8	46.3	93.5	45.2	91.9	92.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	17.8	15.8	50.6	112.2	3.6	1.7	48.0	113.4	45.8	114.4	43.8	115.3	41.2	116.6		
(5)			40.3	85.7	1.8	2.2	39.0	87.3	38.0	88.6	37.0	89.8	35.6	91.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	84.4	75.0	81.3	88.0	93.5	93.3	89.3	83.8	80.8	83.2	86.0	82.4	76.3		
	DBStd	6.82	5.08	5.45	3.96	3.10	3.46	3.90	3.59	2.71	3.05	2.58	3.36	4.30		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	12560	775	877	1179	1305	1342	1178	1049	954	997	1116	971	817		
	CDD65	7086	311	457	714	855	877	728	584	489	547	651	521	352		
	CDH74	95706	3847	6059	10779	14133	14067	10436	6648	4555	5945	8297	6746	4194		
	CDH80	57777	2047	3668	7121	9970	9700	6411	3188	1694	2779	4787	4148	2264		
(14) Wind	WSAvg	7.5	8.2	9.1	9.4	7.6	7.4	7.9	7.5	5.9	5.8	5.9	7.7	8.3		
(15) Precipitation	PrecAvg	19.8	0.0	0.0	0.0	0.3	1.0	2.2	6.1	7.3	3.4	0.8	0.0	0.0		
	PrecMax	30.7	0.0	0.0	0.3	1.9	4.3	4.6	11.9	13.4	9.6	3.2	0.5	0.0		
	PrecMin	8.9	0.0	0.0	0.0	0.0	0.0	0.1	0.7	1.3	0.6	0.0	0.0	0.0		
	PrecStd	5.2	0.0	0.0	0.0	0.4	0.9	1.1	2.4	2.4	1.7	0.9	0.1	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	102.4	107.2	109.6	111.6	111.3	107.4	101.8	94.8	100.0	103.8	103.6	100.6		
		MCWB	64.9	67.4	67.7	69.7	71.8	75.4	78.2	79.8	76.7	70.2	68.0	65.6		
	2%	DB	98.6	104.1	107.5	109.8	109.4	104.4	98.4	91.7	96.7	102.0	101.8	98.1		
		MCWB	63.3	66.3	67.6	70.2	73.0	76.4	78.2	79.2	77.9	69.8	67.1	64.2		
	5%	DB	94.9	101.3	105.6	109.0	107.6	102.5	95.3	89.9	94.6	100.2	100.1	95.1		
		MCWB	61.7	64.7	66.5	70.3	73.3	76.9	77.9	78.8	78.4	69.9	66.4	62.6		
	10%	DB	91.2	97.8	103.5	107.1	105.7	100.3	92.9	88.0	91.8	98.4	98.1	91.8		
		MCWB	59.9	63.0	65.4	70.2	73.7	76.9	78.0	78.3	78.6	71.2	65.1	61.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.9	69.9	76.9	81.0	83.1	83.7	83.9	83.3	83.7	83.2	75.5	68.4		
		MCDB	97.5	102.6	99.4	100.3	100.9	99.2	94.2	90.3	91.9	92.9	92.0	96.3		
	2%	WB	65.3	68.2	71.4	79.1	81.0	82.0	82.1	81.3	81.9	80.7	71.9	66.2		
		MCDB	94.4	101.0	99.8	99.0	99.1	97.2	91.6	87.9	89.9	90.8	92.2	94.2		
	5%	WB	63.0	66.3	68.4	77.3	79.6	80.5	80.5	80.0	80.6	79.0	69.4	64.2		
		MCDB	91.0	97.7	101.4	98.1	97.4	95.2	89.8	86.8	88.8	88.6	93.4	91.3		
	10%	WB	61.0	64.0	66.6	75.5	78.2	79.2	79.1	78.9	79.4	77.8	67.0	62.5		
		MCDB	88.7	94.3	100.0	97.0	95.9	93.3	87.9	85.5	87.6	87.6	92.9	89.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	31.7	31.1	30.1	27.0	23.8	21.1	17.5	14.7	17.9	25.0	31.7	31.6		
		MCDBR	35.2	34.0	32.5	29.6	26.5	24.0	21.0	17.2	21.1	29.0	33.0	34.0		
		MCWBR	15.2	14.8	13.2	12.9	11.4	9.6	9.1	8.3	8.8	11.3	13.6	15.2		
	5% WB	MCDBR	34.2	32.7	30.2	24.0	22.9	21.8	18.7	15.8	18.6	21.9	31.5	33.0		
		MCWBR	16.1	15.3	13.9	12.3	11.4	11.0	10.2	9.3	10.0	10.7	15.1	16.1		
(40) Clear-Sky Solar Irradiance	taub		0.529	0.622	0.692	0.697	0.667	0.661	0.618	0.565	0.566	0.560	0.487	0.488		
	taud		1.821	1.621	1.505	1.543	1.630	1.688	1.814	1.948	1.925	1.888	1.952	1.918		
	Ebn at Noon		245	228	216	214	216	215	225	241	242	241	255	253		
	Edh at Noon		66	84	96	92	82	77	68	61	62	63	57	59		
(44) All-Sky Solar Radiation	RadAvg		1866	1978	2085	2109	2027	1872	1733	1641	1857	1951	1916	1809		
	RadStd		42	55	91	84	105	95	78	81	68	70	50	45		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.74	+1.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page